

Whole Blood is Superior to Component Transfusion for Injured Children

A Propensity Matched Analysis

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Objective: To compare a propensity-matched cohort of injured children receiving conventional blood component transfusion to injured children receiving low-titer group O negative whole blood.

Summary of Background Data: Transfusion of whole blood in pediatric trauma patients is feasible and safe. Effectiveness has not been evaluated.

Methods: Injured children ≥ 1 years old can receive up to 40 mL/kg of cold-stored, uncrossmatched whole blood during initial hemostatic resuscitation. Whole blood recipients (2016–2019) were compared to a propensity-matched cohort who received at least 1 uncrossmatched red blood cell unit in the trauma bay (2013–2016). Cohorts were matched for age, hypotension, traumatic brain injury, injury mechanism, and need for emergent surgery. Outcomes included time to resolution of base deficit, product volumes transfused, and INR after resuscitation.

Results: Twenty-eight children who received whole blood were matched to 28 children who received components. The whole blood group had faster time to resolution of base deficit [median (IQR) 2 (1–2.5) hours vs 6 (2–24) hours, respectively; $P < 0.001$]. The post-transfusion INR was decreased in whole blood vs component cohort [median (IQR) 1.4 (1.3–1.5) vs 1.6 (1.4–2.2); $P = 0.01$]. Lower plasma volumes [median (IQR) = 5 (0–15) mL/kg vs 11 (5–35) mL/kg; $P = 0.04$] and lower platelet volumes [median (IQR) = 0 (0–2) vs 3 (0–8); $P = 0.03$] were administered to the whole blood group versus component group. Other clinical variables (in-hospital death, hospital length of stay, intensive care unit length of stay, and ventilator days) did not differ between groups.

Conclusions: Compared to component transfusion, whole blood transfusion results in faster resolution of shock, lower post-transfusion INR, and decreased component product transfusion. Larger cohorts are required to support these findings.

Keywords: coagulopathy, pediatric, transfusion, trauma, whole blood

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Trauma is the leading cause of death in children, adolescents, and young adults in the United States; hemorrhagic shock after trauma is the most common cause of preventable death.¹ Children in hemorrhagic shock who are hypotensive upon presentation to the emergency department (ED) have a nearly 13-fold increased odds of in-hospital mortality compared to those who are normotensive,² therefore, the early recognition of impending circulatory collapse and aggressive hemostatic resuscitation are critical. Balanced resuscitation with conventional components is the goal in many institutions for the initial trauma resuscitation,^{3,4} as high ratio resuscitation

has been associated with improved hemostasis and survival^{5–8} in adults. There is little data to support the effectiveness of this practice in pediatric trauma resuscitation; in a contemporary civilian pediatric trauma cohort, children requiring massive transfusion have a greater than 50% mortality rate.⁹

Currently, our institutional guideline recommends the transfusion of up to 40 mL/kg of uncrossmatched, leukocyte reduced, low titer (<50) anti-A and -B group O cold-stored whole blood in children age ≥ 1 year with signs of shock after injury as the initial resuscitative fluid. Recent analysis of a pediatric trauma cohort demonstrated an excellent safety profile; there were no adverse transfusion reactions, and hemolytic parameters were not worse amongst the non-group O recipients compared to the group O recipients.¹⁰ These preliminary findings are concordant with the serological safety data in adult trauma patients.^{11–13}

Low titer group O cold-stored whole blood, hereafter referred to as whole blood, offers several advantages over conventional component therapy. Logistically, whole blood transfusion is an efficient alternative to component therapy, requiring less time to obtain and administer a single bag to patients who often have limited intravenous access.¹⁰ Further, overall use of blood products and donor exposures may be reduced with whole blood transfusion. In a single center comparison of whole blood versus component therapy in adult trauma patients, there was a 53% reduction in post-ED blood product transfusion compared to component therapy after controlling for age, injury severity, and prehospital physiology.¹⁴ Biologically, whole blood has a higher hematocrit, greater platelet and factor concentrations, and less additive solution compared to an equivalent volume of reconstituted whole blood. Finally, cold stored platelets have superior function on in vitro coagulation testing compared to room-temperature platelets,^{15–17} which might confer an advantage in the acutely bleeding trauma patient. Cold storage also permits a long storage time and facilitates the utilization of a platelet-containing blood product in prehospital settings and other locations where room-temperature platelets would not otherwise be available.^{18,19}

Barriers to a large study comparing patient outcomes after whole blood transfusion versus conventional component transfusion in injured children include the fact that the use of whole blood is a low frequency event in pediatric trauma, and that this patient cohort is diverse in age, injury mechanism and characteristics. Propensity score matching is a statistical technique that attempts to reduce the bias inherent in an observational study where randomization is not possible. In this study, propensity score matching identified a cohort of critically-injured pediatric patients who received whole blood that was comparable on several clinical covariates to a cohort that did not receive whole blood during their initial resuscitation. The outcomes of this propensity match study included time to resolution of base deficit, volume of blood products transfused, and post-transfusion international normalized ratio (INR) between whole blood versus conventional component pediatric trauma recipients.

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METHODS

Whole Blood Collection and Storage

Rh negative whole blood was procured from a local FDA licensed blood center. The whole blood donors were all male to mitigate the risk of transfusion related acute lung injury and had low titers (<50) of both anti-A and anti-B. The units were collected using a platelet-sparing inline leukoreduction filter (Terumo Imuflex WB-SP, Lakewood, CO) into a bag containing citratephosphate-dextrose solution. Leukoreduction facilitates the recovery of a leukoreduced red blood cell (RBC) from an unused whole blood unit so as to reduce wastage. The whole blood units were stored for up to 14 days between 1–6°C in monitored refrigerators in the ED and in the blood bank.

Low Titer Group O Whole Blood Transfusion Protocol

In June 2016, Low titer group O cold-stored whole blood was approved by the transfusion committee for use in injured pediatric patients, replacing conventional uncrossmatched RBC units as the preferred initial resuscitation fluid at a tertiary care, level 1 pediatric trauma center. The initial eligibility criteria for resuscitation with whole blood at that time included age ≥ 3 years and weight ≥ 15 kg; these criteria were modified in 2018 to age ≥ 2 years and weight ≥ 10 kg, then further modified to age ≥ 1 year in 2019 based on clinical and laboratory safety data. Initially the maximum whole blood dose was 20 mL/kg, which was increased to 40 mL/kg in 2019. During all time periods, the intention during the initial resuscitation was to use the maximum dose of whole blood followed by further resuscitation with conventional component products as indicated by the results of serial rapid thromboelastography testing. Injured patients not meeting the above criteria received conventional component therapy throughout their resuscitation.

Patient Data Collection for Propensity Analysis

Patient data were collected for the propensity analysis in 2 parts. First, the trauma database was queried to identify injured children (age <18 years) who presented to the hospital between January 2013 and May 2016 and received a transfusion of at least 1 blood product in the ED. These patients formed the conventional components group. Second, the trauma database was queried to identify injured children age <18 years who received whole blood during their initial resuscitation between June 2016 and May 2019, the time period after the practice change where whole blood replaced RBC as the recommended initial trauma resuscitation product. These patients formed the whole blood group.

The prospectively-collected data in the database included basic patient demographics, the volumes of all blood products administered, injury severity score and the component abbreviated injury scale scores, admission vital signs, and relevant laboratory test results. All of these trauma patients had conventional coagulation tests, complete blood count, and blood gas analysis performed as part of the standard laboratory panel performed at the time of arrival in the trauma bay. Severe traumatic brain injury (TBI) was defined as head abbreviated injury scale ≥ 3 . Massive transfusion was defined as the transfusion of more than 40 mL/kg of total blood products within a 24-hour period.²⁰

Statistical Analysis

Data for the propensity match were summarized as mean $\pm$ standard deviation, median [interquartile range (IQR)], counts or percentage. Primary outcomes were time to resolution of base deficit, product volumes transfused, and incidence of persistent post-transfusion coagulation dysregulation as defined by the highest INR in the

first 48 hours after admission. Patients with prolonged prehospital cardiac arrest who were subsequently pronounced in the trauma bay were excluded from this analysis. Patients with pre-existing coagulation disorders and those who received component products before arrival at our institution were also excluded. Additional clinical outcomes included functional disability at discharge (defined by a functional independence measure score of dependent in any category or discharge to a rehabilitation facility), hospital length of stay (LOS), intensive care unit (ICU) LOS, time receiving mechanical ventilation, and total transfusion volume.

Propensity-score matching was used to identify a cohort of patients with similar baseline characteristics that were known contributors to trauma-induced coagulopathy, transfusion requirement and poor outcome in pediatric trauma patients: age, mechanism of injury, severe TBI, admission hypotension (age-adjusted), and need for urgent surgery (within 24 hours). Nearest neighbor 1:1 propensity score matching without replacement using a logit model was performed (StataCorp) (1). Standardized differences and standardized percent bias were used to assess the balance of covariates used in propensity score estimation after matching. After propensity matching, variables and outcomes between the 2 groups were compared by Kruskal-Wallis, χ^2 and sign rank tests for categorical and continuous variables. A subgroup analysis was performed to evaluate time to resolution of abnormal admission base deficit; patients were included if they had an abnormal pre-transfusion base deficit (≥ 6) and a post-transfusion base deficit. The time to normalization of the base deficit, calculated by subtracting the time of the first normal post-transfusion base deficit from the time of the first blood product transfusion, was calculated and compared between whole blood and conventional component groups.

Analyses were 2-sided where applicable and a $P < 0.05$ was considered significant. All analyses were performed using Stata version 15 (StataCorp; College Station, TX). Research procedures and analyses conducted for this study were approved by the local institutional review board.

RESULTS

Propensity Analysis

There were 185 injured children who received at least 1 blood product in the ED; 153 who received conventional components between January 2013 and May 2016, and 32 who received whole blood after June 2016. Three children were excluded for preexisting coagulation disorders and twenty-one children were excluded for transfusion before arrival at our institution, all from the component cohort. Four patients who received whole blood and 5 patients who received conventional component transfusion were excluded from the analysis due to being pronounced dead on arrival. Thus, 28 whole blood recipients were matched to 28 conventional component patients who did not receive whole blood (Table 1). There were no significant differences in patient or injury characteristics between these groups, and there was improvement in prematch imbalance across all variables (Fig. 1).

The whole blood group had faster time to resolution of base deficit [median (IQR) 2 (1–2.5) hours vs 6 (2–24) hours, respectively; $P < 0.001$] (Fig. 2). Post-transfusion INR was increased in the component cohort as compared to the whole blood cohort highest INR [median (IQR) 1.6 (1.4–2.2) vs 1.4 (1.3–1.5), respectively; $P = 0.01$]. After adjusting for body weight, lower RBC, plasma and platelet volumes were administered to the whole blood group as compared to the component group (Table 3). There was no difference in the total volume of product transfused, nor in the proportion of patients who were massively transfused in the first 24 hours after injury between the whole blood and conventional components groups

TABLE 1. Patient and Injury Demographics for the LTOWB Recipients Versus the Propensity-matched Blood Product Component Recipients

	LTOWB	Components	P-value
Patient and injury characteristics			
Age (yr)	11 (5–14)	8 (3–13)	0.87
Sex			
Female	9/28 (32%)	11/28 (39%)	0.78
Male	19/28 (68%)	17/28 (61%)	
Race/ethnicity			
White	20/28 (72%)	21/28 (75%)	1.00
Black	5/28 (17%)	5/28 (18%)	
Other/Not specified	3/28 (11%)	2/28 (7%)	
Mechanism of injury			
Blunt	20/28 (72%)	22/28 (78%)	0.87
Penetrating	7/28 (25%)	5/28 (18%)	
Abusive	1/28 (4%)	1/28 (4%)	
Admission GCS median (IQR)	3 (3–15)	5 (3–15)	0.72
Injury Severity Score median (IQR)	27 (19–38)	31 (20–35)	0.46

GCS indicates Glasgow Coma Scale; IQR, interquartile range; LTOWB, low titer group O whole blood.

(48% vs 52%, respectively; $P = 1.00$). For whole blood recipients, there was no difference in the median weight-adjusted volume (mL/kg) of whole blood transfused when comparing time periods of differing maximum allowable volume protocols. Regarding additional clinical outcomes, there was no difference in rate of in-hospital death, functional disability, hospital LOS, ICU LOS, and ventilator days between the 2 groups (Table 2; all $P > 0.25$).

DISCUSSION

This study compared a propensity-matched cohort of injured pediatric patients who received uncrossmatched, low titer group O cold-stored whole blood to those who received conventional components as part of their initial hemostatic resuscitation. Compared to component transfusion, whole blood transfusion resulted in faster resolution of base deficit, a small but statistically significant reduction in the post-transfusion INR, and decreased plasma and platelet transfusion volumes. There were no differences between groups in in-hospital mortality, functional disability, hospital LOS, ICU LOS,

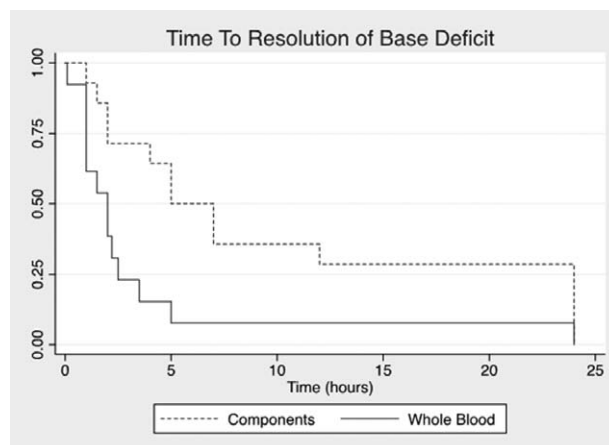


FIGURE 2. Time to resolution of base deficit was decreased in recipients of whole blood as compared to recipients of component products.

TABLE 2. Outcome Measures Between LTOWB and Component Recipients

	LTOWB	Components	P-value
Outcomes			
Mortality	8/28 (29%)	12/28 (43%)	0.40
Functional disability (survivors)	12/21 (57%)	8/16 (50%)	0.75
Hospital length of stay median (IQR)	8 (1–18)	8 (4–13)	0.74
ICU length of stay median (IQR)	4 (2–7)	3 (1–9)	0.89
Ventilator days median (IQR)	2 (0–5)	1 (0–6)	0.8

ICU indicates intensive care unit; IQR, interquartile range; LTOWB, low titer group O whole blood.

or ventilator days between the whole blood and conventional components groups.

Whole blood transfusion for the management of hemorrhagic shock has a lengthy history in military medicine; however, civilian transfusion practice has shifted in the last century toward crystalloid resuscitation and transfusion of component products. More recently, the importance of the early provision of blood products and superiority of products to crystalloids in both military and civilian trauma situations has been documented.^{21–28} Many current protocols for hemostatic resuscitation target a balanced transfusion strategy with ratios of components that attempt to mimic or “reconstitute” the composition of whole blood.^{3,4} With recent advances in transfusion

TABLE 3. Volumes of Blood Products Transfused in LTOWB and Component Cohorts

	LTOWB	Components	P-value
Product volume mL/kg median (IQR)			
Whole blood	15 (10–22)	—	—
Red blood cells	15 (0–28)	24 (10–62)	0.01
Plasma	5 (0–15)	11 (5–35)	0.04
Platelets	0 (0–2)	3 (0–8)	0.03
Total	29 (11–55)	48 (17–122)	0.08

LTOWB indicates low titer group O whole blood.

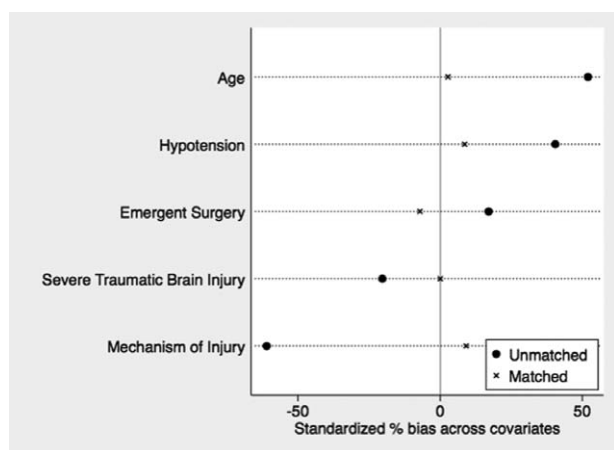


FIGURE 1. Standardized percent bias across covariates was reduced between pre- and post-match cohorts.

practice, whole blood has re-emerged as the standard of care fluid for the initial resuscitation in more than 70 centers across the United States, though in novel and more practical form: cold-stored, low titer anti-A and -B group O whole blood. One limitation to implementing a whole blood program is operational; restocking and recycling product can be both labor- and time-intensive, requiring an active and committed blood bank. Further, some have concerns regarding product waste and cost; the ability to recycle whole blood units to RBC units can alleviate these concerns.

Two recent observational studies reported a mortality benefit for adult patients receiving whole blood as compared to those receiving component products. In a single center comparison of adult trauma patients by Williams et al, whole blood was associated with a 53% reduction in post-ED blood product transfusion and 2-fold increase in likelihood of 28-day survival compared to component therapy after controlling for age, injury severity, and prehospital physiology.¹⁴ A study from Shea et al reported an adjusted mortality benefit at 24 hours and 28 days for adult patients requiring massive transfusion who received whole blood as compared to those who received only component products.²⁹ In observational studies of children to date, no mortality benefit has been seen between whole blood and component groups.^{10,30} This is not surprising; the pediatric cohort needed to demonstrate a survival benefit in a randomized-controlled trial is prohibitively large due to low mortality rates, the current paucity of pediatric centers with whole blood programs, and the relatively small volume of whole blood that had been transfused to the recipients in prior studies. The number of pediatric centers developing whole blood programs is increasing^{31,32} and future multi-center trials may clarify the question of a survival benefit in children. In the propensity-matched study presented here, again, no mortality benefit was seen. This may be attributable to the fact that the majority of deaths were due to TBI as opposed to hemorrhage. Alternately, the volume of whole blood transfused in this study was median (IQR) = 15 (10–22) mL/kg, roughly the equivalent of 1.5 (1–2) units in adults, which may not be a quantity sufficient to yield a mortality benefit.

The use of whole blood affords logistic advantages including efficiency of administration and portability, and the physiologic advantages that we have shown in this report. The earlier clearance of acidosis suggests a more rapid resolution of shock, and improved tissue perfusion. Although the improvement in the INR was modest, elevation of INR has been shown to be one of the strongest predictors of mortality in severely injured children. The lower INR could reflect prevention or more efficient treatment of trauma-induced coagulopathy, though without mechanistic data the significance of this laboratory finding is uncertain; nevertheless, administration of whole blood seems to result in a lower INR than component therapy. At this time, the low numbers of children treated with whole blood, and the relatively low dose of the product administered initially, preclude any conclusions regarding a mortality benefit. The physiologic improvements and other advantages seen in whole blood recipients compared to component recipients are likely multifactorial, though further study is required to evaluate precise mechanism.

Limitations

The main limitation of the study was the relatively small number of whole blood recipients to date. There are limitations inherent in the propensity-matching technique, including the potential for bias due to unobserved cofounders not accounted for in the match. The comparison cohort was a propensity-matched control that was necessarily selected from the time period before whole blood became the standard of care at this hospital, so changes in resuscitation protocols other than the implementation of whole blood could have affected the outcomes. This is unlikely to have been a major factor given the similar in-hospital mortality outcomes in both groups

and lack of major practice changes during that time. Furthermore, data collection regarding transfusion volumes for these subjects was performed retrospectively from medical record review and some inaccuracies due to charting errors might have occurred but would also likely affect both groups of patients. With the exception of admission bloodwork, laboratory testing is performed at the discretion of the provider, therefore the timing is not standardized. Thromboelastography was implemented as the standard of care in 2015, therefore subjects in the historical cohort did not have sufficient data for statistical comparison of functional coagulation parameters. Lastly, these results may not be generalizable to all populations as subjects were drawn from a single academic pediatric institution.

CONCLUSIONS

Compared to conventional blood component transfusion, whole blood transfusion resulted in faster resolution of base deficit, less coagulopathy following transfusion, and decreased plasma and platelet transfusion volumes. Clinical outcomes, including in-hospital mortality, hospital LOS, ICU LOS, and ventilator days, did not differ between propensity-matched subjects who received whole blood during their resuscitation compared to subjects who received conventional component therapy. The expansion of whole blood programs nationally and internationally will facilitate further study regarding the effectiveness of whole blood in preventing and treating trauma-induced coagulopathy and improving patient outcomes.

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